

The University of Nottingham
Malaysia Campus

SCHOOL OF COMPUTER SCIENCE

A LEVEL 3 MODULE, AUTUMN SEMESTER 2012-2013

COMPUTER VISION

Time allowed TWO Hours

Candidates must NOT start writing their answers until told to do so

Answer THREE out of FIVE questions

No calculators are permitted in this examination.

Dictionaries are not allowed with one exception. Those whose first language is not English may use a standard translation dictionary to translate between that language and English provided that neither language is the subject of this examination. Subject specific translation dictionaries are not permitted.

No electronic devices capable of storing and retrieving text, including electronic dictionaries, may be used.

DO NOT turn examination paper over until instructed to do so

1. Answer all of the following sub-questions pertaining to Segmentation (25 marks):

- (a) What is an ill-posed problem? Is segmentation an ill-posed problem? Justify your answer.

[3 marks]

- (b) Consider the set of numbers S below. Apply k-means clustering with $k=1$, $k=2$ and $k=3$. Write down all of your intermediate calculations. What is the best choice of k ? Justify your answer.

[6 marks]

$$S = \{1, 1, 5, 6, 7\}$$

- (c) Describe the split and merge approach to image segmentation. Don't forget to explain the main weaknesses and strengths of the approach.

[5 marks]

- (d) Explain the parallels between the region growing approach to image segmentation and k-means clustering.

[4 marks]

- (e) Consider the 2D matrix of grayscale values below. Convert the matrix into a graph such that it may be used to explain the concept of graph cuts for image segmentation. Draw three distinct cuts on your graph and compute their weights. Relate the weight of these cuts to the quality of the segmentation they represent. Use the cuts in order to motivate the concept of normalized cuts.

[7 marks]

10	12	50	55
15	12	53	55
20	20	55	60
25	25	60	70

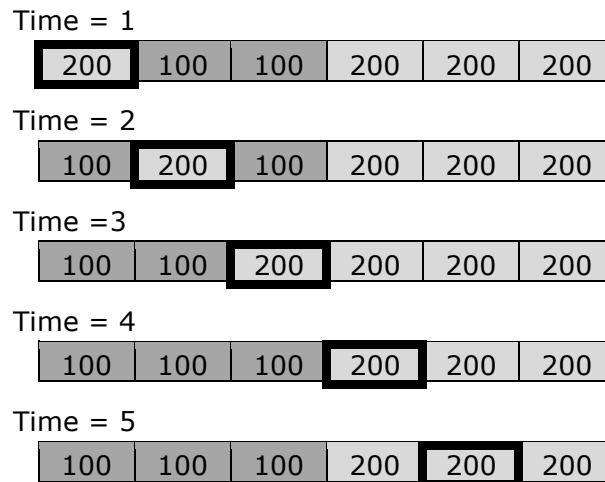
2. Answer all of the following sub-questions pertaining to Motion Analysis (25 marks):

- (a) What are the benefits of motion analysis? What is it used for?

[3 marks]

- (b) Consider the simplified scenario below where an object consisting of a single pixel with grayscale intensity 200 (highlighted with thick borders) is moving from left to right in a one-dimensional "image" consisting of 6 pixels. Use this example to explain how the "frame differencing" approach works. In particular use this example to highlight and explain some of the weaknesses of the approach.

[5 marks]



- (c) What assumptions does the computation of optical flow tend to rely on? Why are these assumptions important? Consider the expression below, which is used for computing horizontal and vertical velocity components. What is the parameter lambda for? If we increase the value of lambda from 0.000001 to 0.001, what effect will this have on the resulting optical flow? Justify your answer.

[5 marks]

$$u = \bar{u} - \lambda \frac{\partial I}{\partial x} \frac{\bar{u} \frac{\partial I}{\partial x} + \bar{v} \frac{\partial I}{\partial y} + \frac{\partial I}{\partial t}}{1 + \lambda \left(\left(\frac{\partial I}{\partial x} \right)^2 + \left(\frac{\partial I}{\partial y} \right)^2 \right)}$$

$$v = \bar{v} - \lambda \frac{\partial I}{\partial y} \frac{\bar{u} \frac{\partial I}{\partial x} + \bar{v} \frac{\partial I}{\partial y} + \frac{\partial I}{\partial t}}{1 + \lambda \left(\left(\frac{\partial I}{\partial x} \right)^2 + \left(\frac{\partial I}{\partial y} \right)^2 \right)}$$

- (d) Consider the problem of detecting squares in images. Assume that the squares can come in different orientations and sizes. Design a simple active contour approach for solving this problem. Describe and explain the main components of your solution.

[6 marks]

- (e) Explain how the condensation algorithm is used for solving the problem of motion-based tracking.

[6 marks]

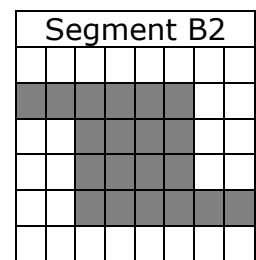
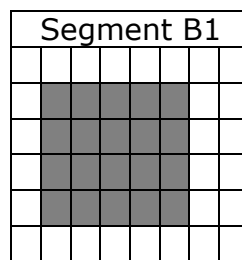
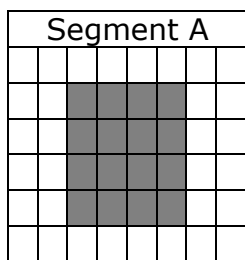
3. Answer all of the following sub-questions pertaining mostly to the Evaluation of Computer Vision algorithms (25 marks):

(a) Describe three different ways for evaluating optical flow algorithms.

[3 marks]

(b) Consider the three segments below (in dark grey). Consider the "Earth mover" distance for computing the distance between segments B1 and A, and segments B2 and A. Consider another distance metric of your choice and compute the distances between the same pairs of segments (i.e. distance(B1,A) and distance(B2,A)). Discuss the pros and cons of the two different distance measures.

[5 marks]



(c) Why are shared test benchmarks important to the goal of algorithm evaluation? When can benchmarks be a problem?

[6 marks]

(d) In the context of optical flow, describe four different ways you might use to compare two velocity vectors, in order to compute some error measure. Include a diagram in your explanation.

[4 marks]

(e) Consider Hidden Markov Models (HMM) for scene analysis. Part of the application of HMM involves learning several parameters. Explain the nature of these parameters. Describe a scene analysis scenario and a possible HMM solution to it. Discuss several issues that might arise at the learning stage.

[7 marks]

4. Answer all of the following sub-questions pertaining to Scene Analysis (25 marks):

- (a) List and briefly explain three problems that Hidden Markov Models generally tackle.
[3 marks]
- (b) What is "hidden" in a Hidden Markov Model, and why is it so? Give an example of a problem where this "hidden" aspect is useful. Why is it useful?
[5 marks]
- (c) How can Hidden Markov Models be used for abnormal behaviour detection?
[6 marks]
- (d) Consider a high-way scenario where cars can exhibit several observable behaviours including: driving at constant speed, accelerating, decelerating, stopping and erratic rotations. Provide a high-level design of a computer vision system capable of detecting car crashes. Don't forget to incorporate all necessary components (e.g. segmentation) and the relationships between these components.
[6 marks]
- (e) Consider the game of badminton. Propose a simple way to apply Hidden Markov Models to the recognition of different badminton strokes (e.g. service, backhand, smash, drop-shot).
[5 marks]

5. Answer all of the following sub-questions pertaining to Stereo Vision (25 marks):

- (a) Explain the relationship between disparity and depth. [4 marks]
- (b) Use low-level features (e.g. grayscale values) to explain some of the main problems underlying the correspondence problem. Include a diagram in your explanation. [5 marks]
- (c) Is camera calibration required for stereo vision? Justify your answer. Give examples of parameters that define different camera setups. [4 marks]
- (d) What is the epipolar line, and what are its implications? In general terms, what is needed in order to compute it? [5 marks]
- (e) Recall the multi-scale solution to optical flow given in the lectures. Describe how you might adapt it to a multi-scale solution to binocular stereo. Give as complete a description of your solution as possible. Include pseudo-code and diagrams if helpful. Discuss potential strengths and weaknesses and justify all points. [7 marks]